

Lecture 1:

Bio informatics a brief History:

Avery-Macleod -McCarty Experiment:

It was an experiment carried out by these three scientists in 1944. That discored that DNA (Deoxyribose Nucleic Acid) is responsible for transferring genes.

Background:

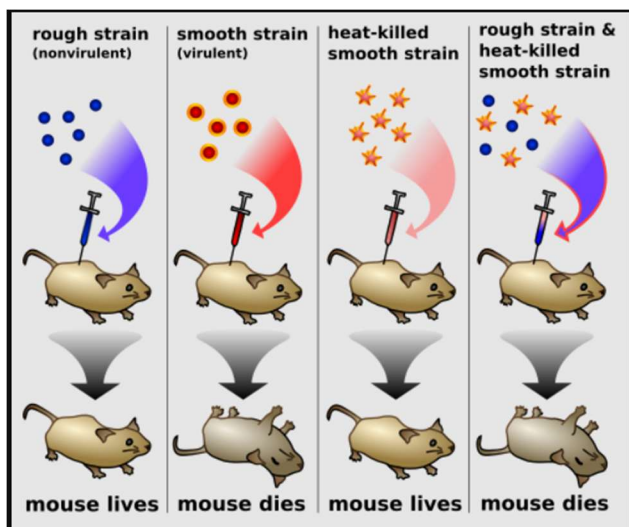
Before this experiment scientists used to think that its protein that is actually responsible for transferring genes rather than DNA because of following reasons:

Protein has more complex structures that suited them well as a gene carrier.

They think DNA is just a merely structural component of chromosomes.

Proteins are way chemically complex than DNA.

Experimental Process:



Extraction Phase:

They first kill the bacteria and treated it with chloroform because protein are insoluble in chloroform and this help isolate the transforming substance.

Purification Phase:

They treated it with alcohol precipitation to make sure it is pure.

Analysis Phase:

They use Trypsin and Chymotrypsin (which breaks down proteins) resulting having no effect on process

Ribonuclease (separating RNA) again no effect

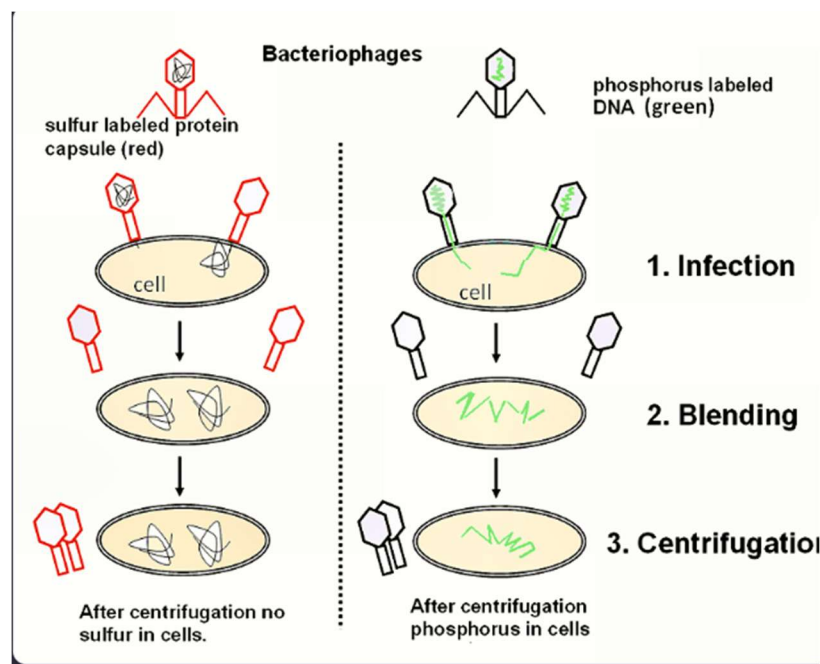
Deoxyribonuclease (separating DNA) inactivated transforming agent

Which shows DNA is transforming agent, but scientist still stuck to protein theory and said that there might be some protein residue left in sample.

But Hershey and Chase experiment in 1952 provide evidence that DNA is the transforming agent.

• Hershey & Chase experiment (1952):

https://www.youtube.com/watch?v=ZtSfFqghEIY&ab_channel=YourekaScience



Hershey and Chase choose simplest organism Bacteriophage A virus that infect bacteria looks like a lunar robot with the protein covering outside and in a shell DNA inside. It

attached to the host or bacteria and inject the dna inside the host while outer covering remain outside the cell.

Now Harshey and chase took the virus and allow them to multiply in a Sulphur 35 radioactive material since only proteins have Sulphur and not DNA so only outer covering of DNA gets radioactive and not the dna inside and then this newly formed virus allowed to attach with bacteria and it inject the DNA in it hence DNA was not radioactive so the newly formed virus inside cell was also not radioactive showing that protein doesn't transfer anything.

Then the allowed virus to replicate in phosphorus 32 radioactive container which makes DNA inside virus radioactive and when it allows to attached with bacteria the virus inside cell is also radioactive showing DNA IS RESPONSIBLE FOR TRANSFERING GENES002E

The Discovery of DNA's Structure (1953)

The following year, James Watson and Francis Crick revealed DNA's three-dimensional structure at Cambridge University. Their model explained how DNA could serve as genetic material:

1. The Double Helix Model

- Two complementary strands twisted together
- Sugar-phosphate backbone on the outside
- Nitrogenous bases paired in the center
- Specific base pairing: adenine (A) with thymine (T), guanine (G) with cytosine (C)

Immediate Impact

- Provided conclusive evidence that DNA is the genetic material
- Explained how genetic information could be stored and transmitted
- Established the foundation for molecular biology

But Rosalind Franklin was the first that discovered the structure of DNA, and was sabotaged by her colleague .

Rosalind Franklin

- Took the crucial X-ray crystallography photograph known as "Photograph 51"
- Her data was instrumental in determining the double helix structure
- Worked at King's College London
- Her contributions were often overlooked until recent years

Margret Dyhoff is considered first Bioinformatician. Her work includes

Atlas of protein sequence and structure

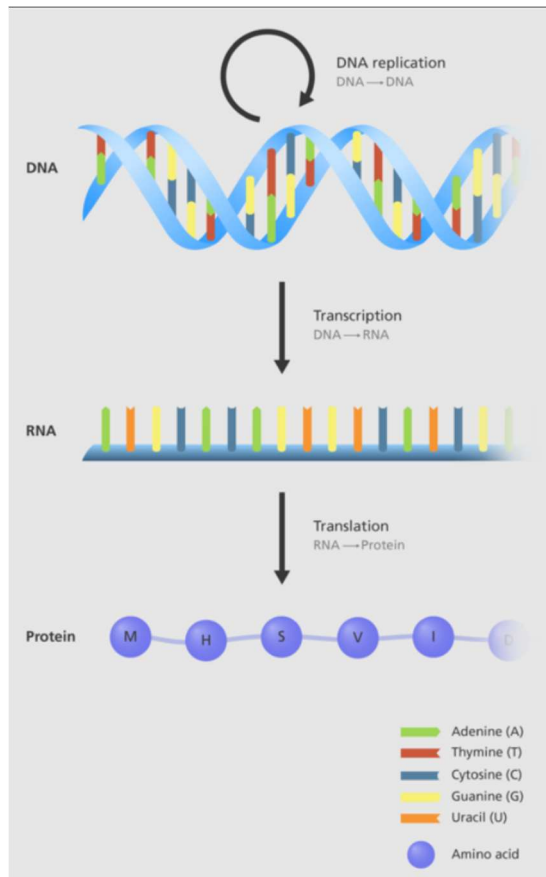
Development of one letter amino acid code

Comproteins first bioinformatics software.

Lecture 2:

Central Dogma:

It is a concept that how actually information is transfer in a body



It has three parts we can say:

DNA:

- Contains all the information about our body (genetic information)
- Present in nucleus of our cell
- It's like a blueprint

RNA:

- Act as a messenger.

- Transfer specific details from DNA
- Contains information that it extract from DNA to make proteins.

Proteins:

- Actual building block of body.
- Performs function and determine traits like body or hair colours.
- Proteins are manufactured in Ribosome

How It Works Together

Think of it like a recipe:

- DNA is like the cookbook with all recipes
- RNA is like a photocopy of one specific recipe
- Proteins are like the finished dishes

Real-World Example

For example, if you have red hair:

1. The instructions for red hair are stored in your DNA
2. When your body needs to make the protein that creates red hair color
3. RNA carries those specific instructions from DNA
4. The protein-making machinery uses those instructions to create the protein that gives your hair its red color

What is Metabolism?

Metabolism is like a conveyor belt or manufacturing line in body that makes protein.

Think of your body like a factory with three main components:

1. Genes: These are like blueprints or instructions
2. Enzymes: These are like specialized tools that do specific jobs

3. Metabolism: This is like the factory's production line

How It All Works Together

1. Genes → Enzymes

- Genes contain instructions for making enzymes
- Each gene is like a recipe for a specific enzyme
- When a gene works correctly, it produces a working enzyme

2. Enzymes → Metabolism

- Enzymes are biological catalysts
- They help break down or build up substances in your body
- Each enzyme has a specific job, like a specialized worker

What Happens in Metabolic Diseases

When there's a problem:

1. Gene Problem

- The gene's instructions are incorrect or incomplete
- This is called a **mutation**
- The gene can't produce a working enzyme

2. Enzyme Problem

- The enzyme isn't made or doesn't work properly
- The specific job that enzyme does isn't getting done
- This causes problems in metabolism

3. Disease Result

- The metabolic pathway gets blocked
- Substances can build up or not get processed
- This leads to disease symptoms

This understanding of genes, enzymes, and metabolism has led to many treatments for metabolic diseases. By knowing exactly which enzyme is missing or not working, doctors can often provide specific treatments to help manage the condition.

DNA:

Structure

DNA has a unique double helix structure that was first described by Watson and Crick in 1953

. This structure consists of:

1. The Double Helix

- a. Two complementary strands twisted together
- b. Strands are opposite direction i.e ($5' \rightarrow 3'$ and $3' \leftarrow 5'$) indicate the direction of each strand

2. The Building Blocks

- Sugar molecules (deoxyribose)
- Phosphate groups
- Four nitrogenous bases:
 - Adenine (A)

- Guanine (G)
- Cytosine (C)
- Thymine (T)

3. Pairing Rules:

Adenine always pairs with Thymine, and Guanine always pairs with Cytosine

Function

DNA serves several critical functions in cells:

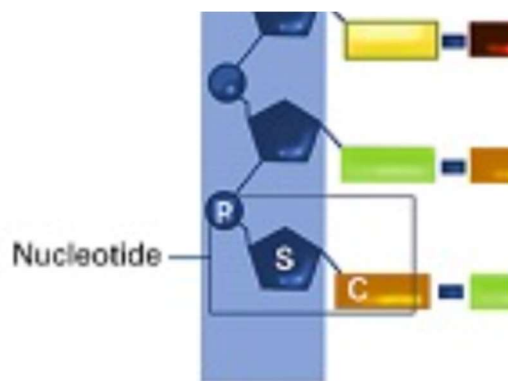
1. Genetic Information Storage

- Contains instructions for making proteins
- Stores genetic information in a compact form
- Each human cell contains about 2 meters of DNA packed into a tiny nucleus

2. Protein Synthesis

- Provides instructions through the Central Dogma:
 - DNA → RNA → Proteins
- Each gene (section of DNA) codes for a specific protein
- The human genome contains around 30,000 genes

Nucleotides:



Nucleotides are the fundamental building blocks of DNA and RNA, serving as the basic units that store genetic information. Let's break down their components and structure in detail.

Basic Structure

A nucleotide consists of three main components:

1. Sugar Molecule

- Deoxyribose in DNA
- Ribose in RNA
- Forms the backbone of the molecule

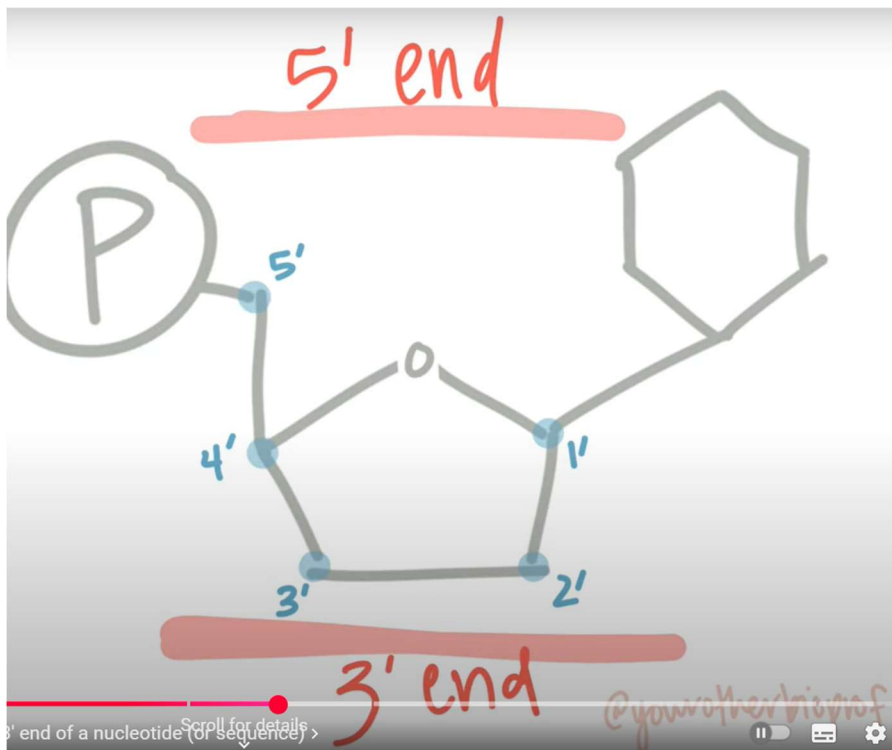
2. Phosphate Group

- Links sugar molecules together
- Forms strong bonds between nucleotides

3. Nitrogenous Base

- DNA: Adenine (A), Guanine (G), Cytosine (C), Thymine (T)
- RNA: Adenine (A), Guanine (G), Cytosine (C), Uracil (U)
- Contains the genetic information

5' and 3':



5' and 3' are basically location or you can say name of specific carbon atom in nucleotide

Think of DNA like a one-way street:

- 5' (five prime) is the starting point
- 3' (three prime) is the ending point
- DNA processes only work in one direction: from 5' to 3'

Just like you can only drive one way on a one-way street, DNA can only be built or read in the 5' to 3' direction. This direction is crucial for DNA synthesis and genetic information processing

Purines and pyrimidines?

Nitrogenous bases can be further classified as pyrimidines or purines. Cytosine, uracil, and thymine are all pyrimidines. That is, their molecular structure comprises a nitrogenous base in the form of a six-member single ring.

Guanine and adenine, on the other hand, are purines. These contain a nitrogenous base in the form of a nine-member double ring. In short, pyrimidines have only one ring while purines have two.

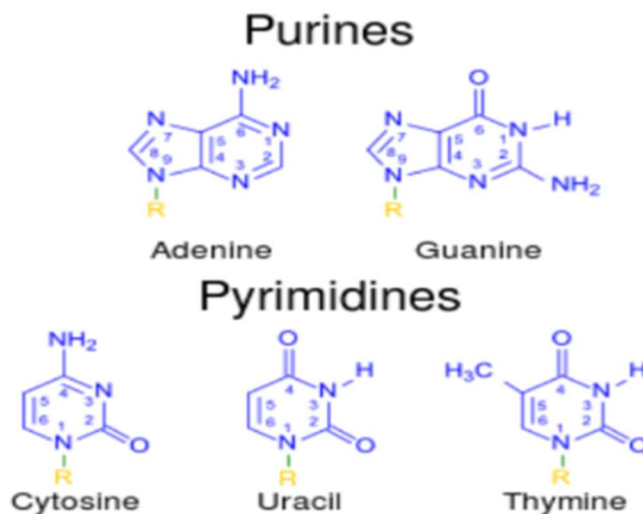


Image Source: Wikimedia Commons

RNA (Ribonucleic acid)

It is a crucial molecule that plays several vital roles in protein synthesis and cellular function. Let's explore its structure and types in detail.

Basic Structure

RNA has several key characteristics:

- Single-stranded molecule (unlike DNA's double helix)
- Contains ribose sugar (instead of deoxyribose) which mean one less oxygen atom in Sugar molecule.
- Uses uracil instead of thymine
- Generally smaller than DNA molecules

Types of RNA

1. Messenger RNA (mRNA)



- Acts as a blueprint for protein synthesis (copies information from DNA)
- Copied from DNA in the nucleus
- Contains codons that specify amino acids(Codons are three pair of nitrogenous bases (A,T,G,C))
- Short-lived and specific to each protein

2. Transfer RNA (tRNA)

- Has a distinctive cloverleaf shape
- Carries specific amino acids to the ribosome
- Contains anticodons that match mRNA codons
- Each tRNA is specific to one amino acid

3. Ribosomal RNA (rRNA)

- Makes up a large part of ribosomes
- Forms the site where proteins are assembled
- Helps position tRNA molecules correctly

- Essential for protein synthesis

4. Small nuclear RNA (snRNA)

- Involved in RNA splicing
- Helps remove introns (non-coding regions)
- Essential for mRNA processing

RNA Polymerases

There are three main types of RNA polymerase in bacteria:

1. Polymerase I

- Makes most types of RNA
- Involved in rRNA synthesis
- Essential for cell survival

2. Polymerase II

- Primarily makes mRNA
- Involved in transcription regulation
- Can be inhibited by certain antibiotics

3. Polymerase III

- Makes tRNA and some small RNAs
- Important for protein synthesis
- Highly processive enzyme

Codon-Anticodon System

The genetic code is read in groups of three bases (codons):

- mRNA codons are read in sequence
- Each codon specifies one amino acid
- tRNA anticodons match mRNA codons
- This system ensures accurate protein synthesis (manufacturing of protein)

For example, the codon AUG in mRNA matches with the anticodon UAC in tRNA, which brings the amino acid methionine to the growing protein chain. This precise matching system ensures that proteins are built correctly.

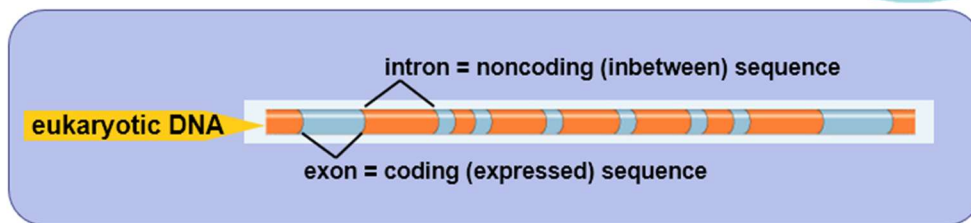
Exons and Introns:

Eukaryotic genes have junk!

- Eukaryotic genes are not continuous

- ◆ **exons** = the real gene
 - **expressed** / coding DNA
- ◆ **introns** = the junk
 - **inbetween** sequence

introns
come out!



AP Biology

mRNA's require EDITING before use

- Message is NOT CONTINUOUS
- INTRONS are removed

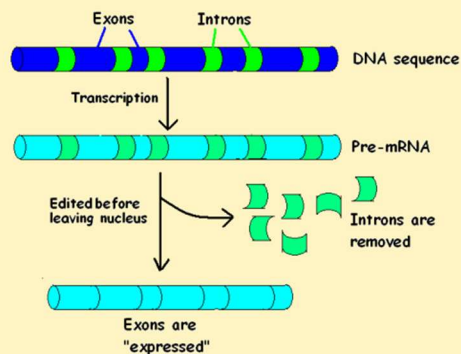


Image by Riedell

Transcription:

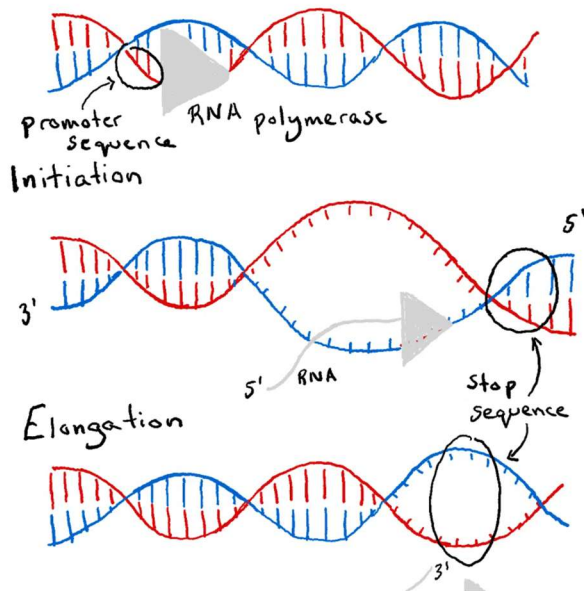
Overview

Transcription is the process in which genetic code from DNA is transferred into RNA.

Pre-transcription Preparation

Before transcription begins, the cell prepares in three ways:

Step	Process	Purpose
DNA Unwinding	→ Double helix opens up	→ Makes DNA readable
Strand Selection	→ Cell chooses template strand	→ Ensures correct genetic information
RNA Polymerase Binding	→ Enzyme attaches to DNA	→ Starts transcription process



Main Transcription Process

Transcription happens in three main stages

1. Initiation

- RNA polymerase binds to DNA
- DNA unwinds at specific region
- Process begins at "promoter" site

2. Elongation

- RNA polymerase reads DNA template
- Matches nucleotides (A-T, G-C, U replaces T)
- Builds complementary RNA strand

3. Termination

- Process reaches end signal
- RNA polymerase releases
- New RNA molecule is complete

Post-transcription Processing

After transcription, the RNA goes through important modifications:

| RNA Splicing → Removes unwanted segments (introns) |

| Cap Addition → Protects RNA from breakdown (5' CAP)

| Tail Addition → Stabilizes RNA molecule (poly A Tail)

Key Enzymes and Proteins

Enzyme/Protein	Role
RNA Polymerase	Main enzyme that reads DNA and builds RNA
Helicase	Unwinds DNA double helix
Spliceosomes	Remove introns during splicing
Endonucleases	Cut RNA at specific points

Alternative Splicing

Cells can create different RNA products from the same DNA sequence through alternative splicing. This helps create variety in proteins produced from a single gene, allowing cells to adapt to different situations.

Translation:

Translation is the process by which cells convert messenger RNA (mRNA) into proteins. Let me explain this complex process in detail, focusing on the key components and their roles.

Core Components

1. Ribosome

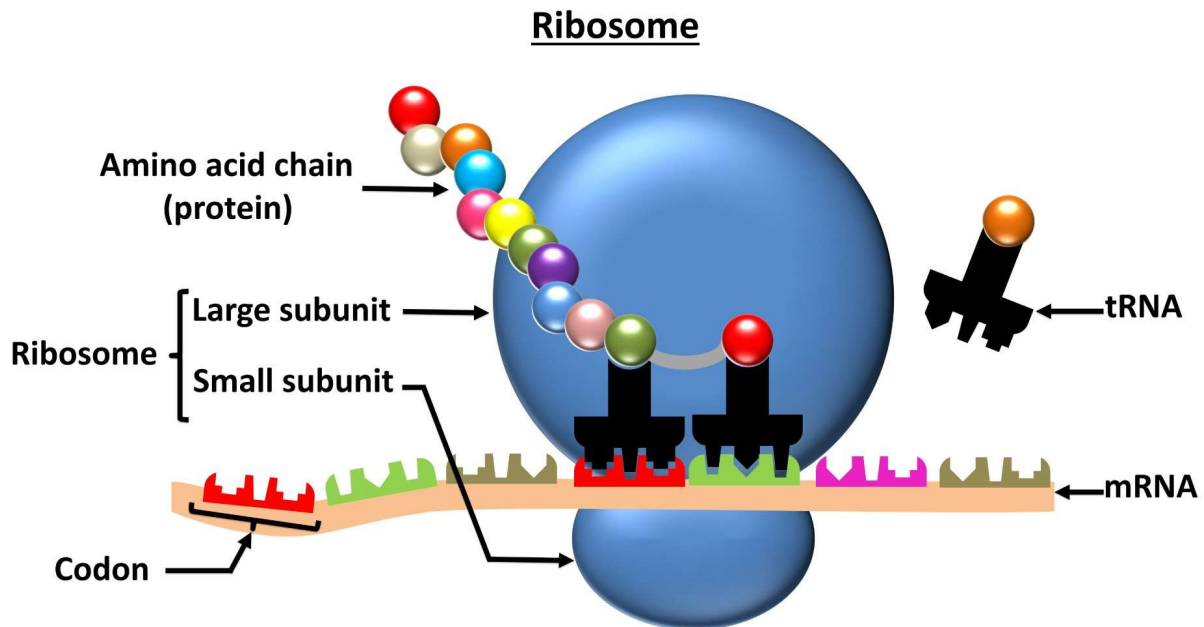
- Complex cellular machine that reads mRNA and builds proteins 3:3
- Made of two subunits: large (50S) and small (30S)
- Contains special RNA molecules (rRNA) and proteins
- Has three binding sites: A (amino acid), P (polypeptide), and E (exit)

2. Transfer RNA (tRNA)

- Acts as a bridge between mRNA and amino acids
- Has two important ends:
- Anticodon: reads mRNA codons
- Attachment site: carries amino acids
- Each tRNA recognizes specific codons through complementary base pairing

Translation Process

Translation occurs in three main phases:



1. Initiation Phase

- Small ribosomal subunit binds to mRNA
- Starts at AUG codon (start signal)
- First tRNA brings **methionine** (always the first amino acid)
- Large ribosomal subunit joins to complete the complex

2. Elongation Phase

- tRNAs bring amino acids to the ribosome
- Process follows three-step cycle:
 - tRNA binding to A site
 - Peptide bond formation
 - Ribosome translocation along mRNA

3. Termination Phase

- Process ends at stop codons (UAA, UAG, UGA)
- No tRNA recognizes stop codons
- Release factors trigger completion
- Final protein is released

Peptide Bond Formation

Peptide bonds are the chemical links that connect amino acids into a protein chain:

1. Bond Creation

- Formed between amino acids in adjacent positions
- Creates strong covalent bonds
- Links amino acids end-to-end
- Process catalyzed by ribosomal RNA itself

2. Polypeptide Chain Growth

- Chain grows one amino acid at a time
- New amino acids add to C-terminal end
- Previous amino acids remain attached
- Chain continues until stop codon reached

Important Terms Explained

Term	Definition
Codon	Three-nucleotide sequence in mRNA that codes for an amino acid
Anticodon	Complementary sequence on tRNA that matches codon
Polypeptide	Long chain of amino acids linked by peptide bonds
Peptidyl Transferase	Activity that catalyzes peptide bond formation
Initiation Factors	Proteins that help start translation process
Elongation Factors	Proteins that facilitate translation continuation

This complex process ensures accurate protein synthesis, with multiple checkpoints and mechanisms to guarantee proper function. The precise coordination between ribosomes, tRNAs, and mRNA allows cells to build proteins essential for life.

Mutation:

[youtube.com](https://www.youtube.com)

A mutation is a change in the sequence of nucleic acids (DNA or RNA) that can occur in any living organism, from bacteria to humans

. These changes can happen randomly or be caused by various factors, and they're essential for evolution and genetic diversity.

Types of Mutations

1. Point Mutations

- **Definition:** Changes in a single nucleotide base
- **Types**
 - Silent mutations: No change in amino acid produced
 - Missense mutations: Different amino acid produced
 - Nonsense mutations: Creates premature stop codon

2. Frameshift Mutations

- **Definition:** Insertions or deletions of nucleotides
- **Effect:** Changes the reading frame of genetic code
- **Example**

Original: AUG-AAU-ACG-GCU

After insertion: AUG-AAA-UAC-GGC-U

Chromosomal Mutations

- **Types:**
 - Duplication: Extra copies of genes
 - Deletion: Loss of genetic material
 - Inversion: Reversed chromosome segment
 - Translocation: Fragment moves to different chromosome

Real-World Example: Sickle Cell Anemia

A classic example of a point mutation is sickle cell anemia

- Caused by a single nucleotide substitution in hemoglobin gene
- Results in abnormal hemoglobin protein
- Affects red blood cell shape and oxygen transport
- Can be inherited from parents

Mutation Effects

1. **Neutral:** Many mutations have no effect on the organism
2. **Beneficial:** Some mutations provide advantages (e.g., sickle cell trait offers malaria protection)
3. **Harmful:** Can lead to genetic disorders or reduced function

Eukaryotic and prokaryotic:

Basic Cell Structure

Feature	Eukaryotic Cells	Prokaryotic Cells
Nucleus	Has true nucleus with membrane	No true nucleus
Organelles	Has membrane-bound organelles	Lacks membrane-bound organelles
Genetic Material	Linear chromosomes in nucleus	Single circular chromosome
Cell Wall	Present	Present
Cytoplasm	Complex with many organelles	Simple with few structures

Key Differences in Cellular Organization

1. Nuclear Organization

- Eukaryotes: DNA contained within nucleus
- Prokaryotes: DNA floats freely in cytoplasm

2. Organelle Structure

- Eukaryotes: Complex organelles with membranes
- Prokaryotes: Simple structures without membranes

3. Cellular Complexity

- Eukaryotes: Highly organized with specialized compartments
- Prokaryotes: Simpler organization with fewer specialized structures

Gene expression:

Gene expression is the process by which the information in a gene's DNA is converted into a functional product, such as a protein, that can affect the organism. Let me explain this complex process in detail.

Basic Process Overview

Gene expression follows a specific sequence of steps that convert genetic information into a functional product. Here's how it works:

1. Transcription

- DNA is copied into messenger RNA (mRNA)
- Occurs in the nucleus (eukaryotes) or cytoplasm (prokaryotes)
- Creates complementary RNA copy of gene

2. Translation

- mRNA is read by ribosomes
- Amino acids are assembled into proteins
- Occurs in cytoplasm

Reading Frames

A reading frame is the sequence of three nucleotides (triplets) that cells use to read genetic information. Think of it like reading a sentence three letters at a time:

Original: THEFATCATSATSUN

Reading: THE-FAT-CAT-SAT-SUN

In genetics, each triplet (called a codon) specifies one amino acid or a stop signal .

Open Reading Frames (ORFs)

An open reading frame is a sequence of nucleotides that:

1. Starts with a start codon (AUG)
2. Ends with a stop codon (UAA, UAG, or UGA)
3. Contains a sequence of codons that can be translated into a protein

Dna Replication:

DNA replication is the process by which cells make an exact copy of their DNA before cell division. This complex process ensures that each new cell receives a complete and accurate set of genetic instructions.

Basic Process Overview

DNA replication follows a specific sequence of steps that ensure accurate copying of genetic material. Here's how it works:

1. Initiation

- DNA unwinds at specific regions called origins
- Enzyme helicase breaks hydrogen bonds between strands
- Replication fork forms where DNA separates

2. Unwinding

- DNA double helix opens up
- Topoisomerase enzyme helps relieve tension
- Single-strand binding proteins stabilize separated strands

3. Synthesis

- DNA polymerase reads template strands
- New nucleotides added in 5' to 3' direction
- Continuous on leading strand, discontinuous on lagging strand

Key Components

COMPONENT	FUNCTION
Helicase	Unwinds DNA double helix
DNA Polymerase	Adds new nucleotides to growing strand
Ligase	Seals gaps between nucleotides
Primase	Creates RNA primers
Topoisomerase	Relieves DNA tension
Single-strand binding proteins	Stabilizes separated strands

Leading vs Lagging Strand Synthesis

1. Leading Strand

- Synthesized continuously
- New nucleotides added in 5' to 3' direction
- Single long piece of DNA

2. Lagging Strand

- Synthesized discontinuously
- Forms short segments called Okazaki fragments

- Each fragment about 1000-2000 nucleotides long

TATA box:

The TATA box is a special DNA sequence that helps control how genes are turned on and off in cells. It's like a signpost that tells the cell where to start reading genetic instruction.

What is it?

The TATA box is a short piece of DNA that contains repeating T and A bases. It's usually found about 25-35 base pairs before where genes start being read

How does it work?

The TATA box works by:

1. Helping cells find where to start reading genes
2. Guiding special proteins to the right spot
3. Helping turn genes on and off when needed

SNP:

A Single Nucleotide Polymorphism (SNP) is a variation in a single nucleotide (A, T, C, or G) that occurs at a specific position in the DNA sequence among individuals of the same species. Think of it like a single letter difference in a word between two people's copies of the same book.

Basic Concept

SNPs are the most common type of genetic variation in humans. They occur when there's a difference in a single DNA building block at a particular position in the genome. For example:

Person A: ATCGTTA

Person B: ATCCTTA

Key Features

Feature	Description
Location	Can occur anywhere in the genome
Frequency	Common in human population
Effect	May or may not affect gene function
Inheritance	Passed from parents to children
Variation	Usually involves two different nucleotides

Types of SNPs

1. Coding Region SNPs

- Occur in genes
- May affect protein function
- Can be silent or cause disease

2. Non-coding SNPs

- Found in regulatory regions
- Can affect gene expression
- Often associated with disease risk

Clinical Significance

1. Disease Association

- Help identify genetic disorders
- Used in personalized medicine
- Guide treatment decisions

2. Research Applications

- Understanding disease mechanisms
- Developing new treatments
- Improving genetic testing

RECAP:

[quick recap](#)

Amino Acids:

Amino acids are the building blocks of proteins, and they come in 20 different standard types that combine to create all proteins in living organisms. Each amino acid has unique properties that determine how proteins function and what they look like.

Basic Structure

All amino acids share a common structure with:

- A central carbon atom (called the alpha carbon)
- An amino group (-NH₂)
- A carboxyl group (-COOH)
- A hydrogen atom
- A unique side chain (R group)

Properties and Classification

Property	Description
Hydrophobic	Water-repelling (nonpolar)
Hydrophilic	Water-attracting (polar)
Acidic	Contains negative charge

Basic	Contains positive charge
Essential	Must be obtained from diet
Non-essential	Can be made by the body

Essential Amino Acids

These nine amino acids must come from our diet:

1. Histidine (His) - Basic, important for protein structure
2. Isoleucine (Ile) - Hydrophobic, branched chain
3. Leucine (Leu) - Hydrophobic, branched chain
4. Lysine (Lys) - Basic, important for protein structure
5. Methionine (Met) - Hydrophobic, contains sulfur
6. Phenylalanine (Phe) - Hydrophobic, aromatic
7. Threonine (Thr) - Polar, important for protein structure
8. Tryptophan (Trp) - Hydrophobic, aromatic
9. Valine (Val) - Hydrophobic, branched chain

Non-essential Amino Acids

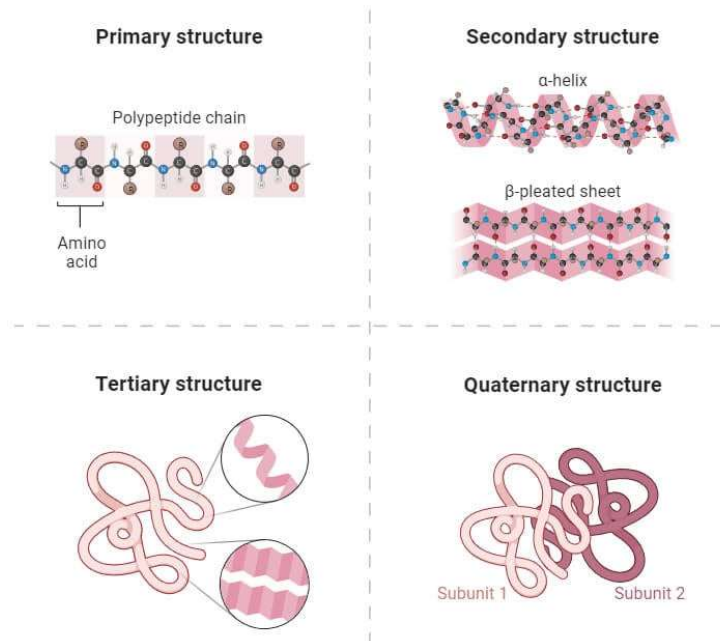
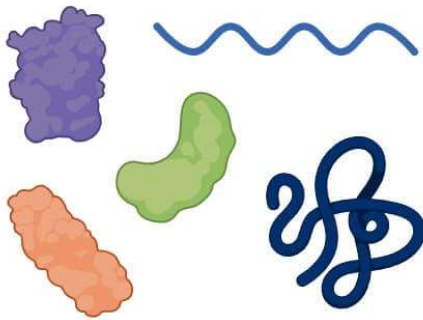
These can be made by the body:

1. Alanine (Ala) - Hydrophobic, small
2. Arginine (Arg) - Basic, important for protein structure
3. Asparagine (Asn) - Polar, uncharged
4. Aspartic acid (Asp) - Acidic
5. Cysteine (Cys) - Polar, contains sulfur
6. Glutamic acid (Glu) - Acidic
7. Glutamine (Gln) - Polar, uncharged

8. Glycine (Gly) - Nonpolar, smallest amino acid
9. Proline (Pro) - Hydrophobic, cyclic structure
10. Serine (Ser) - Polar, uncharged
11. Tyrosine (Tyr) - Polar, aromatic

Four level of protein structure: [youtube](#)

Protein Structure



1. Primary Structure

primary protein structure = sequence of amino acids ⁱ

Ala-Gly-Val-Tyr-Arg-Leu-Ser-Met-Asn-Cys-Pro

The **primary structure** of a protein is the **linear sequence of amino acids** in a polypeptide chain. These amino acids are linked together by **peptide bonds**.

- This sequence is determined by the **DNA code** through the process of **transcription and translation**.

- Even a small change in the amino acid sequence (like a mutation) can affect the protein's function.
- Example: In **sickle cell anemia**, just one amino acid change in hemoglobin leads to the disease.

2. Secondary Structure

The **secondary structure** refers to the **local folding or coiling** of the polypeptide chain due to **hydrogen bonding** between nearby amino acids.

The two most common secondary structures are:

- **Alpha helix (α -helix)** – a spiral shape stabilized by hydrogen bonds.
- **Beta pleated sheet (β -sheet)** – a folded sheet-like structure where strands run side-by-side.

These structures provide initial **stability and shape** to the protein.

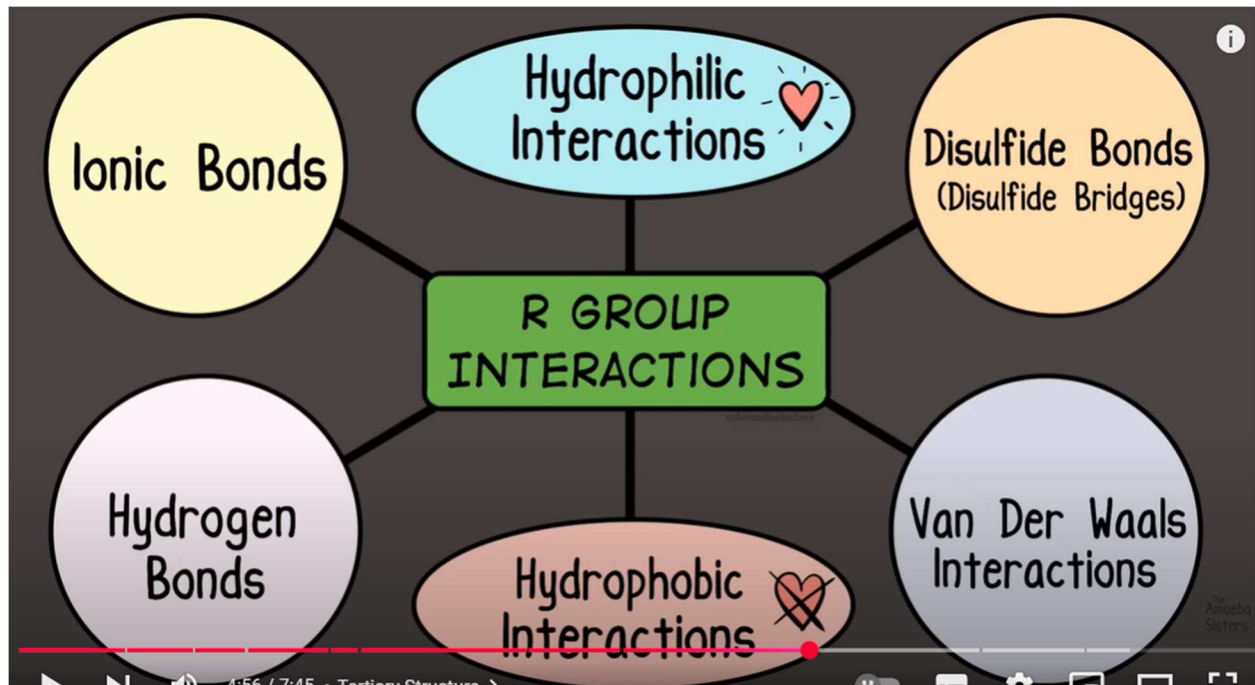
3. Tertiary Structure

The **tertiary structure** is the **overall 3D shape** of a single polypeptide chain. It is formed by the interactions between the **side chains (R groups)** of amino acids.

These interactions include:

- **Hydrophobic interactions** (non-polar side chains group away from water)
- **Hydrogen bonds**
- **Ionic bonds**
- **Disulfide bridges** (covalent bonds between two cysteine amino acids)

This level determines the **biological activity** of the protein. If the protein is **misfolded**, it may lose its function or cause diseases.



4. Quaternary Structure

The **quaternary structure** exists only in proteins that have **more than one polypeptide chain**. These multiple chains, called **subunits**, come together to form a functional protein.

- Example: **Hemoglobin** has four subunits (two alpha and two beta chains).
- The arrangement and interaction of subunits influence the protein's function and stability.

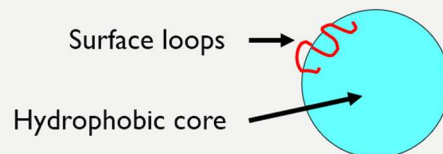
How Protein Structure and Function Are Related

The **structure of a protein determines its function**—this is one of the central ideas in biology. Proteins are made of amino acids, and the way these amino acids fold and interact gives the protein its unique shape. That shape is what allows the protein to do its job.

Protein Structure

Protein core - usually conserved.

Protein loops - variable regions



PROTEINS – THE MACHINES IN YOU

To keep you alive, your body must do many things...

- Convert oxygen to usable energy
- Digest food and recover nutrients
- Build and maintain body tissue
- Fight off invasions and detoxify poisons
- Take out the garbage

Proteins do all of these things!

And when proteins do not work correctly...

- Disease
- Deformities
- Death

X-ray Crystallography

What is it?

X-ray crystallography is a technique used to determine the **3D structure of molecules**—especially large ones like proteins and DNA—by analyzing how **X-rays scatter** when they hit a **crystal of the molecule**.

How does it work?

1. **Crystallization:** First, the protein is purified and turned into a solid crystal. This step can be difficult and time-consuming.

2. **X-ray exposure:** The crystal is bombarded with **X-ray beams**.
3. **Diffraction pattern:** As X-rays hit the crystal, they scatter and form a pattern.
4. **Data analysis:** A computer analyzes the diffraction pattern to calculate the **electron density map**.
5. **Model building:** Scientists use the map to build a **3D model** of the molecule.

Advantages

- Can produce **very high-resolution** 3D structures.
- Suitable for **large and complex** molecules.

Limitations

- Requires **crystallizing** the protein, which is hard for some proteins.
- Cannot capture molecules in **dynamic or flexible states**—only the static structure in the crystal.



NMR (Nuclear Magnetic Resonance) Spectroscopy

What is it?

NMR spectroscopy is a technique that uses **magnetic fields and radio waves** to study the structure of molecules in **solution** (like water). It's like MRI for molecules.

How does it work?

1. The protein is placed in a **strong magnetic field**.
2. Radio waves are used to excite the **nuclei of atoms** (especially hydrogen).
3. These nuclei emit signals that depend on their **chemical environment**.
4. The signals are analyzed to reveal the **distances between atoms** and how the molecule is folded.

Advantages

- No need for crystals; works in **solution**, closer to real biological conditions.
- Can show **dynamic changes** in structure (e.g., protein folding).
- Great for **small to medium-sized** proteins (usually < 40 kDa).

Limitations

- Less suitable for **large proteins**.
- Produces lower resolution structures compared to X-ray crystallography.
- Requires a lot of **data and time** for complex molecules.